PC 604 CS	DEEP LEARNING					
Pre-requisites			**L**	**T**	**P**	**C**
			3	-	-	3
Evaluation	**SEE**	60 Marks	**CIE**		40 Marks	

Course Objectives :	
1	To introduce basic concepts of artificial neural networks and multilayer perceptrons
2	To introduce basic concepts of CNN and VGG
3	To introduce recurrent neural networks and LSTM's
4	To introduce auto encoders and GAN's

Course Outcomes :	
On completion of this course, the student will be able to :	
CO-1	Understand the problem of XOR seperability and activation functions in ANN's
CO-2	Understand the problem of over fitting, under fitting, Gradient Descent and Stochastic Gradient Descet
CO-3	Demonstrate understanding of CNN's and VGG's
CO-4	Demonstrate understanding of RNN's and LSTM's
CO-5	Use auto encoders and GAN's

UNIT – I

Artificial Neural Networks: Introduction, Perceptron, XOR Gate, Perceptron Training Rule, Activation Functions

Linear Neural Networks: Linear Regression, Implementation of Linear Regression, Softmax Regression, The Image Classification Dataset , Implementation of Softmax Regression

UNIT – II

Multilayer Perceptrons: Multilayer Perceptrons, Implementation of Multilayer Perceptrons, Model Selection, Under fitting and Over fitting, Weight Decay, Dropout, Forward Propagation, Backward Propagation, and Computational Graphs, Numerical Stability and Initialization, Considering the Environment, Predicting House Prices.

Optimization Algorithms: Optimization and Deep Learning, Convexity, Gradient Descent, Stochastic Gradient Descet, Minibatch Stochastic Gradient Descent, Momentum, Adagrad, RMSProp, Adadelta, Adam, Learning Rate Scheduling

UNIT – III

Introduction to Convolutional Neural Networks: Introduction to CNNs, Kernel filter, Principles behind CNNs, Multiple Filters

Modern Convolutional Neural Networks: Deep Convolutional Neural Networks (AlexNet), Networks Using Blocks (VGG), Network in Network (NiN), Networks with Parallel Concatenations (GoogLeNet), Batch Normalization, Residual Networks (ResNet), Densely Connected Networks (DenseNet)

UNIT – IV

Recurrent Neural Networks: Sequence Models, Text Preprocessing, Language Models and the Dataset, Recurrent Neural Networks, Implementation of Recurrent Neural Networks from Scratch, Concise Implementation of Recurrent Neural Networks, Back propagation Through Time.

Modern Recurrent Neural Networks: Gated Recurrent Units (GRU),Long Short Term Memory (LST), Deep Recurrent Neural Networks, Bidirectional Recurrent Neural Networks, Machine Translation and the Dataset, Encoder-Decoder Architecture, Sequence to Sequence, Beam Search

UNIT –V

Auto encoders: Types of Auto Encoders and its applications

Generative Adversarial Networks: Generative Adversarial Network, Deep Convolutional Generative Adversarial Networks

Suggested Reading:

1	Goodfellow, I., Bengio,Y., and Courville, A., Deep Learning, MIT Press, 2016. Link:https://www.deeplearningbook.org
2	Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, Dive into Deep Learning, 2020
3	Dive into Deep Learning — Dive into Deep Learning 0.16.6 documentation (d2l.ai)